1. Object Detection Performance Comparison

Two object detection models, Viola–Jones (Model A) and YOLO (Model B), are compared using their True Positive (TP), False Positive (FP), and False Negative (FN) counts.

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A (Viola–Jones)	80	20	30
Model B (YOLO)	90	15	20

(a) Precision and Recall Calculation

Formulas used:

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

Model A (Viola–Jones):

Precision = 80 / (80 + 20) = 80 / 100 = 0.80 (80%)

Recall = 80 / (80 + 30) = 80 / 110 = 0.727 (72.7%)

Model B (YOLO):

Precision = 90 / (90 + 15) = 90 / 105 = 0.857 (85.7%)

Recall = 90 / (90 + 20) = 90 / 110 = 0.818 (81.8%)

Model	Precision	Recall
Model A (Viola–Jones)	0.80 (80%)	0.727 (72.7%)
Model B (YOLO)	0.857 (85.7%)	0.818 (81.8%)

(b) Comparison and Justification

Model B (YOLO) outperforms Model A (Viola–Jones) on both metrics: precision is higher by about 5.7 percentage points (85.7% vs 80%), and recall is higher by about 9.1 percentage points (81.8% vs 72.7%). A higher precision means YOLO produces fewer false alarms relative to its correct detections, while the higher recall means it misses fewer actual objects (fewer false negatives).

Since Model B dominates Model A on both precision and recall simultaneously (rather than trading one off for the other), YOLO (Model B) provides the better overall object detection performance.

2. Motion Estimation Error Analysis

Two motion estimation techniques are evaluated using their displacement errors (in pixels) across three consecutive frames.

Method	Frame 1	Frame 2	Frame 3
Method A	2	3	2
Method B	3	5	4

(a) Average Displacement Error

Average Error = (Sum of errors) / (Number of frames)

Method A:

Average Error = (2 + 3 + 2) / 3 = 7 / 3 = 2.33 pixels

Method B:

Average Error = (3 + 5 + 4) / 3 = 12 / 3 = 4.00 pixels

(b) & (c) Comparison and Justification

Method A has a lower average displacement error (2.33 pixels) compared to Method B (4.00 pixels). Since a smaller displacement error indicates the estimated motion is closer to the true motion between frames, Method A provides more accurate motion estimation than Method B. Method B's errors are also consistently higher in every individual frame (3 vs 2, 5 vs 3, and 4 vs 2), reinforcing that it is the less accurate technique.

3. Depth Estimation Comparison

A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects, compared against the actual (ground-truth) depths.

Object	Actual Depth (m)	Stereo Vision (m)	SfM (m)
Object 1	2.0	2.0	2.5
Object 2	3.0	3.0	3.8
Object 3	4.0	4.0	5.0

(a) Absolute Error for Each Object

Absolute Error = | Estimated Depth − Actual Depth |

Object	Stereo Vision Error (m)	SfM Error (m)
Object 1	$\lvert 2.0 - 2.0 \rvert = 0.0$	$\lvert 2.5 - 2.0 \rvert = 0.5$
Object 2	$\lvert 3.0 - 3.0 \rvert = 0.0$	$\lvert 3.8 - 3.0 \rvert = 0.8$
Object 3	$\lvert 4.0 - 4.0 \rvert = 0.0$	$\lvert 5.0 - 4.0 \rvert = 1.0$

(b) Average Absolute Error

Average Absolute Error = (Sum of absolute errors) / (Number of objects)

Stereo Vision:

Average Error = (0.0 + 0.0 + 0.0) / 3 = 0.0 m

SfM:

Average Error = (0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.767 m

(c) Comparison and Conclusion

The Stereo Vision system produced zero absolute error for all three objects, giving it an average absolute error of 0.0 m, while the SfM system produced an average absolute error of 0.767 m. Since a lower absolute error indicates estimates closer to the true depth values, the Stereo Vision system provides more accurate depth estimation than the Structure-from-Motion (SfM) system for this dataset.